

STAT3003: Survey Methods

5. Systematic Sampling

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5.1 Introduction

- We have covered **three survey sample designs**
 - Simple random sampling
 - Stratified random sampling
 - Cluster sampling
- **Both involve simple random samples** at some level
 - With simple random sampling, it is across the population
 - With stratified random sampling, it is within each stratum
 - With cluster sampling, it is across the primary units
 - With two stage cluster sampling, it is across the primary units and then across the secondary units

5.1 Introduction

- But sometimes actually performing a simple random sample is difficult
 - Imagine you want to survey shoppers outside Langham Place in Mong Kok
 - What's N ?
 - How would you draw up a list of sampling units (shoppers) from which you could randomly select your sample (via lots, random number tables or RAND() in Excel)?

5.1 Introduction

- And sometimes a simple random sample selection may provide a **misleading sample**
 - Imagine you want to estimate the proportion of faulty light bulbs made in a factory in a day
 - Unbeknown to you, part of the machinery makes the light bulbs become loose and produced nothing but faulty light bulbs from 3pm onwards
 - Your simple random sample might contain too many/not enough of these faulty light bulbs

5.1 Introduction

- A solution to problems like these is **systematic sampling**
 - Suppose you want a sample of size n
 - Choose an **interval** and **select names at equal intervals** along the 'list' of names
- So outside Langham Place, select every tenth shopper you see until you have surveyed 200
- In the light bulb factory, select every 20th light bulb for inspection

5.1 Introduction

- A formal definition of systematic sampling is the following:
 - A sample obtained by randomly selecting one element from the first k elements in the frame and every k th element thereafter is called a 1-in- k systematic sample with a random start
- Example
 - Our population consists $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 - Use a 1-in-3 systematic sample with a random start
 - We randomly select one of 1, 2, 3
 - Let's say 3 is chosen
 - Then our sample will be elements $\{3, 6, 9\}$

5.1 Introduction

- We can see how a systematic sample **might be easier to conduct in the field** than a simple random sample (Langham Place)
- **If the population has certain pattern,** systematic sampling **may** offer greater information per unit cost than simple random sampling (light bulb factory)

5.1 Introduction

- You are told that the light bulb factory produces 3000 light bulbs a day
- You want a sample size of at least 300
- Thus, if you want to conduct a 1-in- k systematic sample with a random start, k must be less than or equal to 10 in order for you to achieve your desired sample size

5.1 Introduction

- More generally, say the population size is N , your desired sample size is n
 - Then we must have $k \leq N/n$
- What if we do not know N ?
 - As in the Langham Place survey
- We must guess what k must be to achieve a sample of size n
 - If k is too large, we might not reach the desired sample size

5.2 Estimation under Systematic Sampling

- Surprisingly, systematic sampling is related to cluster sampling.
- Here's a population of 96 elements, which we have labelled

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72
73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88
89	90	91	92	93	94	95	96

5.2 Estimation under Systematic Sampling

- Suppose we conduct a 1-in-5 systematic sample. There are five possible samples, denoted by red, green, yellow, black and white

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72
73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88
89	90	91	92	93	94	95	96

5.2 Estimation under Systematic Sampling

- Say the randomly selected start is element 3. Then our sample would be the elements in yellow

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64
65	66	67	68	69	70	71	72
73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88
89	90	91	92	93	94	95	96

5.2 Estimation under Systematic Sampling

- Essentially, we have partitioned the population into $N = 5$ clusters (primary units), randomly selected $n = 1$ of them and measured every element inside
- So **systematic sampling** $\approx$ **cluster sampling with $n = 1$**

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30
31	32	33	34	35
36	37	38	39	40
41	42	43	44	45
46	47	48	49	50
51	52	53	54	55
56	57	58	59	60
61	62	63	64	65
66	67	68	69	70
71	72	73	74	75
76	77	78	79	80
81	82	83	84	85
86	87	88	89	90
91	92	93	94	95
96				

5.2 Estimation under Systematic Sampling

- The notation is very similar notation to before
 - N is the number of primary units in the population
 - n is the number of primary units in the sample
 - M_i is the number of secondary units in the i th primary unit
 - $M = \sum_{i=1}^N M_i$ is the total number of secondary units in the population
 - Y_{ij} is the j th observation in the i th primary unit
 - $Y_i = \sum_{j=1}^{M_i} Y_{ij}$ is the total of observations in the i th primary unit

5.2 Estimation under Systematic Sampling

- The cluster sample with n clusters sampled estimator for τ is
 - $N \frac{1}{n} \sum_{i=1}^n Y_i = N\bar{Y}$
- So an appropriate estimator for τ under systematic sampling (denoted $\hat{\tau}_{sys}$) is $N \bar{Y}$ with $n = 1$
 - $\hat{\tau}_{sys} = NY_1$
- From the cluster sampling estimator, we know $\hat{\tau}_{sys}$ is unbiased

5.2 Estimation under Systematic Sampling

- Can we use the cluster sampling formulae for the variance?
- For cluster sampling,
 - $Var(\hat{t}) = N^2 \frac{N-n}{N-1} \frac{1}{n} \sigma_c^2$, where σ_c^2 is the variance of the cluster totals
- Hence, for systematic sampling
 - $Var(\hat{t}_{sys}) = N^2 \sigma_c^2$
- For cluster sampling,
 - $\widehat{Var}(\hat{t}) = N^2 \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}_c^2}{n}$, where $\hat{\sigma}_c^2$ is the sample variance of the cluster totals
- Hence for systematic sampling...

5.3 The Problem With Variance

- Clearly, with a sample of $n = 1$ we cannot estimate σ_c^2
 - $\Rightarrow$ we cannot calculate $\widehat{Var}(\hat{\tau})$
 - $\Rightarrow$ we cannot calculate confidence intervals for τ
- Many surveys use a systematic sampling design with $n = 1$. If we cannot calculate a confidence interval for our estimator using cluster sampling formulae, what can we do?
 - Make an assumption about the population

5.3 The Problem With Variance

- If we can treat our sample of the M_1 secondary units in the single primary unit chosen **as if it were a SRS of size M_1 from the $M = \sum_{i=1}^N M_i$ secondary units in the population**, then we could use the SRS formulae
- This is only reasonable if we can assume the secondary units are **in random order**
 - i.e. there is no pattern or structure to the population

5.3 The Problem With Variance

- Therefore, if we think the population of secondary units is indeed randomly ordered, we may use

$$- \text{Var}(\hat{t}_{sys}) \approx \frac{M-M_1}{M-1} \frac{M^2}{M_1} \sigma^2$$

$$- \widehat{\text{Var}}(\hat{t}_{sys}) \approx M^2 \left(1 - \frac{M_1}{M}\right) \frac{1}{M_1} \hat{\sigma}^2$$

$$- \Rightarrow \widehat{\text{Var}}(\hat{t}_{sys}) \approx \frac{M^2}{M_1} \left(1 - \frac{M_1}{M}\right) \frac{1}{M_1-1} \sum_{j=1}^{M_1} (Y_{1j} - \bar{Y}_1)^2$$

- ... to calculate confidence intervals and required sample sizes

5.3 The Problem With Variance

- if we think the secondary units are randomly ordered throughout the population, then **treating the systematic sample as an SRS of size M_1 units** is reasonable
- But what if we think the secondary units are **not** randomly ordered?
- We need to dig deeper to understand when the SRS estimator for variance
 - Underestimates the true variance
 - Reasonably estimates the true variance
 - Overestimates the true variance

5.3 The Problem With Variance

- Recall the variance of cluster totals, σ_c^2
 - $\sigma_c^2 := \frac{1}{N} \sum_{i=1}^N (y_i - \mu_c)^2$ where $\mu_c = \frac{1}{N} \sum_{i=1}^N y_i$
 - Note $\mu_c = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{M_i} y_{ij} = \frac{M}{N} \frac{1}{M} \sum_{i=1}^N \sum_{j=1}^{M_i} y_{ij}$
 - Thus we may write $\mu_c = \bar{M} \mu$, where μ is overall population mean (i.e. the mean of all the secondary units)
- Let us investigate the variance of the cluster totals further
 - We define the **sum of squares** in σ_c^2 to be $\sum_{i=1}^N (y_i - \mu_c)^2$

5.3 The Problem With Variance

- These sum of squares may be written as
$$- \sum_{i=1}^N (y_i - \mu_c)^2 = \sum_{i=1}^N \left[\sum_{j=1}^{M_i} y_{ij} - \bar{M}\mu \right]^2$$
- For simplicity, assume $M_1 = M_2 = \dots = M_N = \bar{M}$ (i.e. the clusters all have the same size.)
$$\begin{aligned} - \Rightarrow \sum_{i=1}^N (y_i - \mu_c)^2 &= \sum_{i=1}^N \left[\sum_{j=1}^{\bar{M}} y_{ij} - \bar{M}\mu \right]^2 \\ &= \sum_{i=1}^N \left[\sum_{j=1}^{\bar{M}} (y_{ij} - \mu) \right]^2 \\ &= \sum_{i=1}^N \left[\sum_{j=1}^{\bar{M}} (y_{ij} - \mu)^2 + \sum_{j=1}^{\bar{M}} \sum_{k \neq j} (y_{ij} - \mu)(y_{ik} - \mu) \right] \end{aligned}$$
- These two terms mean something by themselves.

5.3 The Problem With Variance

- Consider the **population variance of the secondary units**. By definition, this must be

$$- \sigma^2 := \frac{1}{N\bar{M}} \sum_{i=1}^N \sum_{j=1}^{\bar{M}} (y_{ij} - \mu)^2$$

- The **within-primary-unit correlation coefficient** is the correlation from all possible pairs (y_{ij}, y_{ik}) , $j \neq k$ in the population
 - It measures how similar elements in the same cluster are; a measure of homogeneity within the clusters
 - It is also called the intraclass/intraclass correlation coefficient (**ICC**)

5.3 The Problem With Variance

- For cluster i , with size M_i , there are $M_i(M_i - 1)$ possible pairs (y_{ij}, y_{ik}) , $j \neq k$
 - Therefore, across all clusters there are $\sum_{i=1}^N M_i(M_i - 1)$ possible pairs
 - Taking $M_i = \bar{M}$ for $i = 1, \dots, N$, we see there are $N\bar{M}(\bar{M} - 1)$ possible pairs
- Remembering the definition of correlation, we define the ICC to be

$$- ICC := \frac{\left(\sum_{i=1}^N \sum_{j=1}^{\bar{M}} \sum_{j \neq k} (y_{ij} - \mu)(y_{ik} - \mu) \right)}{N\bar{M}(\bar{M} - 1)\sigma^2}$$

5.3 The Problem With Variance

- We now see the sum of squares relationship
 - $\sum_{i=1}^N (y_i - \mu_c)^2 = \sum_{i=1}^N \left[\sum_{j=1}^{\bar{M}} (y_{ij} - \mu)^2 + \sum_{j=1}^{\bar{M}} \sum_{k \neq j} (y_{ij} - \mu)(y_{ik} - \mu) \right]$
 - May be rewritten as
 - $N\sigma_c^2 = N\bar{M}\sigma^2 + N\bar{M}(\bar{M} - 1)\sigma^2 ICC$
- Or, more simply
 - $\sigma_c^2 = \bar{M}\sigma^2(1 + (\bar{M} - 1)ICC)$
 - This equation relates the variance of the primary units (σ_c^2) to that of the secondary units (σ^2)

5.3 The Problem With Variance

- From the cluster sampling chapter, we know that under cluster sampling

$$- \text{Var}(\hat{\tau}) = N^2 \text{Var}(\bar{Y}) = N^2 \frac{N-n}{N-1} \frac{1}{n} \sigma_c^2$$

$$- = N^2 \frac{N-n}{N-1} \frac{\bar{M}}{n} \sigma^2 (1 + (\bar{M} - 1) ICC)$$

- Now imagine that, for the same population, we conducted an **SRS of size $n\bar{M}$ secondary units**. The estimator for τ would have variance

$$- \text{Var}(\hat{\tau}_{SRS}) = \frac{(N\bar{M} - n\bar{M})}{N\bar{M} - 1} \frac{(N\bar{M})^2}{n\bar{M}} \sigma^2 = N^2 \frac{(N-n)}{N-1} \frac{\bar{M}}{n} \frac{N-1}{N - \frac{1}{\bar{M}}} \sigma^2$$

$$- \approx N^2 \frac{N-n}{N-1} \frac{\bar{M}}{n} \sigma^2$$

5.3 The Problem With Variance

- Therefore, we can see
 - $ICC \approx 0 \Rightarrow Var(\hat{t}_{SRS})$ estimates the true variance well
 - $ICC \gg 0 \Rightarrow Var(\hat{t}_{SRS})$ underestimates the true variance
 - $ICC \ll 0 \Rightarrow Var(\hat{t}_{SRS})$ overestimates the true variance
- In the setting of systematic sampling, which kinds of populations will have $ICC \approx 0, \gg 0$ or $\ll 0$?
 - It all depends how the sampling frame is ordered.

5.3 The Problem With Variance

- If the sampling frame is in **random order**, then $ICC \approx 0$
 - Quite often the ordering of the population is unrelated to the characteristic of interest
 - E.g. your characteristic of interest is monthly salary; the frame is ordered by the last 4 digits of their phone number

5.3 The Problem With Variance

- If the sampling frame is in **increasing or decreasing order**, then $ICC < 0$
 - In this case, elements near to each other are more similar than elements that are farther apart
 - **The systematic samples forces the sample values to be spread out**, making the elements within each possible cluster less homogeneous
 - E.g. the debts owed to a bank may be listed chronologically, from shortest maturity to the longest. Longer maturity debts tend to be larger than shorter maturity debts, thus the sampling frame may have an increasing order

5.3 The Problem With Variance

- If the sampling frame has a **periodic pattern**, we need to be careful
 - We must avoid choosing a 1-in- k systematic sample where k coincides with a multiple of the period
 - In this case the values within a cluster will be extremely similar, making $ICC \gg 0$
 - E.g. the sampling frame is 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3 and we choose $k = 3$

5.3 The Problem With Variance

- Example (from Scheaffer et al. 2012)
 - You are the boss of a factory making bronze piping.
 - (Bronze is an alloy of mainly copper and tin)
 - A few of your customers complain to you that the quality of your piping has deteriorated recently
 - You conduct a sample by selecting one pipe after each 30 minutes of production and measuring its % of copper
 - Your minimum standard is 90%
 - You observe an average of 87% and a s.d. of $\sqrt{18}\%$ from a sample of 16 measurements
- Has your product declined in quality?

5.3 The Problem With Variance

- We will use the SRS formulae for standard error. How will compare with the true error variance?
 - Trends in the quality of piping (up or down) are possible, in which case the SRS formulae will overestimate the true standard error
 - It seems unlikely that the quality is periodic with a period of around 30 min
 - Perhaps it is most likely that the quality is randomly ordered throughout the day

5.3 The Problem With Variance

- We are told
 - $\bar{y}_{sys} = 87$ and $s^2 = 18, n = 16$
- Use the 95% C.I. for a SRS (note the normality assumption we have made)
 - $\Rightarrow \left(87 \pm t_{15,0.975} \sqrt{\left(1 - \frac{16}{N}\right) \frac{18}{16}} \right)$
- Taking **N to be large** (again, reasonable)
 - $\Rightarrow \left(87 \pm 2.13 \times \frac{3}{2\sqrt{2}} \right) = (84.74, 89.26)$
- The survey suggests that the quality has declined. You need to find out why.

5.3 The Problem With Variance



- Example
 - The management of a factory wants to estimate p , the proportion of workers who support Tottenham Hotspur F.C.
 - From the $N = 2000$ employees leaving the factory after work, they perform a 1-in-10 systematic sample with a random start
 - They observe 132 workers who support Tottenham. Use this information to estimate p and provide a 95% C.I.

5.4 Repeated Systematic Sampling

- Not being able to estimate $Var(\hat{\tau}_{sys})$ directly has meant that **we have to rely on our systematic sample being very similar to a simple random sample**
- Very often, this is not the case
- Then **we need a different method** to estimate $Var(\hat{\tau}_{sys})$
- We can use repeated systematic sampling

5.4 Repeated Systematic Sampling

- Repeated systematic sampling involves selecting **multiple** systematic samples
- Example: we have a population of size 300, we want a systematic sample of size 60
 - We could use a 1-in-5 systematic sample across the population
 - Or, we could use a 1-in-50 systematic sample with a random start across the population and repeat this 10 times

5.4 Repeated Systematic Sampling

- Both methods produce 60 measurements for estimating μ , say
- Both methods require roughly the same amount of resources
- But the second method produces 10 measurements for estimating $Var(\hat{t}_{sys})$
- This is useful if we suspect that our sample will be quite different from a simple random sample

5.4 Repeated Systematic Sampling

- Let n_s be the number of repeated systematic samples
- We usually require $n_s \geq 10$ in order to have a satisfactory estimate of $Var(\hat{t}_{sys})$
- Let the repeated samples be 1-in- k' systematic samples with random starts
- Let the single sample we also considered be a 1-in- k systematic sample with a random start
- Then we choose k' such that $k' = kn_s$

5.4 Repeated Systematic Sampling

- The **starting points** for each of the n_s repeated systematic samples are **chosen randomly from the first k' elements**
- The other elements in each repeated systematic sample are chosen by adding $k', 2k', 3k'$ and so on until each repeated sample contains n/n_s elements

5.4 Repeated Systematic Sampling

- Example
 - The population contains 1440 elements, we want a systematic sample of size 80
 - Which would require an interval of $k = 18$
 - We suspect that this sample would be quite different from a simple random sample, so we use repeated random sampling
 - We choose $k' = 180 = 10 \times 18 = n_s k$
 - Which means we will perform 10 repeated 1-in-180 systematic samples with random starts, each with sample size $\frac{1440}{180} = 8 = N/k'$

5.4 Repeated Systematic Sampling

- To choose the 10 random starts, we randomly select 10 random numbers from 1 to 180
- Say Excel gives us 73, 42, 81, 145, 6, 21, 86, 17, 112, 102
- Then the elements in each sample would be...

1st (element in sample)	2nd	3rd	4th	5th	6th	7th	8th
73	253	433	613	793	973	1153	1333
42	222	402	582	762	942	1122	1302
81	261	441	621	801	981	1161	1341
145	325	505	685	865	1045	1225	1405
6	186	366	546	726	906	1086	1266
21	201	381	561	741	921	1101	1281
86	266	446	626	806	986	1166	1346
17	197	377	557	737	917	1097	1277
112	292	472	652	832	1012	1192	1372
102	282	462	642	822	1002	1182	1362

5.4 Repeated Systematic Sampling

- Repeated systematic sampling is essentially equivalent to cluster sampling
- Thus all the formulae from cluster sampling apply
 - $\hat{t} = N \frac{1}{n} \sum_{i=1}^n Y_i$
 - $Var(\hat{t}) = \frac{N-n}{N-1} \frac{1}{n} \sigma_c^2$, where σ_c^2 is the variance of cluster totals
 - $\widehat{Var}(\hat{t}) = N(N-n) \frac{1}{n} \hat{\sigma}_c^2$, where $\hat{\sigma}_c^2$ is the sample variance of the cluster totals
 - A $100(1 - \alpha)\%$ is given by $\hat{t} \pm t_{n-1, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{t})}$

5.4 Repeated Systematic Sampling

- Systematic sampling shares the same guiding principle as cluster sampling
 - To obtain estimators with low variance (or MSE), the population should be partitioned into **primary units (clusters) which are similar to each other**
 - The perfect primary unit contains secondary units which are truly representative of the population of secondary units
 - The within-primary-unit variance does not appear in $\widehat{Var}(\hat{t})$, **so we should choose clusters with the highest within-primary-unit variance as possible**

5.4 Repeated Systematic Sampling

- In systematic sampling, the secondary units inside a primary unit are often “spaced apart”
 - E.g. dividing a geographic area into regions (primary units), regions into blocks (secondary units)
 - E.g. dividing shoppers at Langham Place systematically, e.g. 1-in-20
- Therefore, quite often, a systematic sample follows the guiding principle

5.4 Repeated Systematic Sampling

- Whereas, in cluster sampling, which is mainly motivated by convenience or practicality, the guiding principle is often not followed
 - E.g. dividing a geographic area into neighbouring regions, observing every plot inside a region
 - E.g. observing the 10th to 19th shoppers at Langham Place, then the 30th to 39th, etc.
- Thus cluster sampling often provides estimators with a high variance

5.4 Repeated Systematic Sampling

- Example (adapted from Scheaffer et al. (2012))
 - It's 1966 in Hong Kong. There are no cross-harbour tunnels yet.
 - [A ferry carries cars across Victoria Harbour](#). The ferry company charges per car, not per person.
 - The company wants to estimate the average number of people per car for the month of January
 - The company also knows that last January, 400 cars were carried across the harbour.
 - They want to sample 80 cars.

5.4 Repeated Systematic Sampling

- Instead of doing a single 1-in-5 systematic sample, they opt to use repeated systematic sampling
- They select 10 repeated samples with 8 samples in each
 - That is ten 1-in-50 samples
- Thus ten numbers are chosen from $\{1, 2, \dots, 50\}$ by SRS
 - These provide the ten random starts for the 1-in-50 systematic samples

5.4 Repeated Systematic Sampling

- The cars chosen are outlined in this table

Random start	2nd Element	3rd Element	4th Element	5th Element	6th Element	7th Element	8th Element	$\bar{y}_i$
2	52	102	152	202	252	302	352	3.75
5	55	105	155	205	255	305	355	3.375
7	57	107	157	207	257	307	357	2.875
13	63	113	163	213	263	313	363	4.625
26	76	126	176	226	276	326	376	4.5
31	81	131	181	231	281	331	381	5.25
35	85	135	185	235	285	335	385	4.5
40	90	140	190	240	290	340	390	4.125
45	95	145	195	245	295	345	395	4.25
46	96	146	196	246	296	346	396	4.375

5.4 Repeated Systematic Sampling

- What are the primary units? How many are there?
- What are the secondary units? How many are there in a primary unit? How many are there in the total population?
- Estimate the mean number of persons per car
- Provide a 95% C.I.